Key

Notes 0.1: Simplifying Expressions, PEMDAS and Solving Equations **Adv Algebra 2**

Simplify expressions

Re-writing an expression in a simpler, more concise form without changing its value.

Examples:

Simplify each expression.

1. $n + 4 - 9 - 3n$

$-2n - 5$

2. $-4x^2 + 7 + 8x^2 - 4$

$4x^2 + 3$

3. $-2(1 - 9x) + 6(x - 10)$

$-2 + 18x + 6x - 60$

$24x - 62$

Order of operations

PEMDAS stands for what? Parentheses, Exponents, Multiplication, Division, Addition, Subtraction

Examples:

Evaluate each expression.

1. $8^2 \div (2 \cdot 8) + 2$

$8^2 \div (16) + 2$

$64 \div 16 + 2$

$4 + 2$

$\boxed{6}$

2. $3^2 \div 3 + 2^2 \cdot 7 - 20 \div 5$

$9 \div 3 + 4 \cdot 7 - 20 \div 5$

$3 + 28 - 4$

$\boxed{27}$

3. $\frac{4(5^2) - 4 \cdot 3}{4(4 \cdot 5 + 2)}$

$\frac{4(25) - 4 \cdot 3}{4(20+2)}$

$\frac{100 - 12}{4(22)}$

$\frac{88}{88}$

$\boxed{1}$

Solving an equation

To solve an equation, the goal is to isolate the variable (which is an unknown value) on one side of the equal sign.

Examples:

Solve each equation.

1. $75 - 9g = 5(-4 + 2g)$

$75 - 9g = -20 + 10g$

$-10g \quad -10g$

$75 - 19g = -20$

$-75 \quad -75$

$\frac{-19g}{-19} = \frac{-95}{-19}$

$\boxed{g = 5}$

2. $\frac{8}{2x+3} = \frac{12}{x-4}$

$8(x-4) = 12(2x+3)$

$8x - 32 = 24x + 36$

$-16x = 68$

$x = -\frac{68}{16}$

$\boxed{x = -\frac{17}{4}}$

3. $\frac{7}{6}x + 15 = -\frac{1}{3}(2x - 15)$

$\frac{7}{6}x + 15 = -\frac{2x}{3} + 5$

$\frac{7x}{6} + \frac{2x}{3} = -10$

$\frac{7x}{6} + \frac{4x}{6} = -10$

$\frac{11x}{6} = -10$

$x = -60$

$x = -$